



Mining air pollution monitoring network at Las Palmeras coal mine in Colombia

CLIENT

GECELCA S.A. E.S.P.

SECTOR

Mining

LOCATION

Cordoba, Colombia

PRODUCTS

Kunak AIR Pro
Kunak AIR Cloud

Need

Gecelca S.A. E.S.P. (La Generadora y Comercializadora de Energía del Caribe) contacted Ambientalía with the need to create a mining air pollution monitoring system to validate the air quality at Las Palmeras coal mine in Cordoba, Colombia.

In 2020, during the mine's construction phase, a first diagnosis phase was carried out using a mobile solar station for measuring particulate matter (PMs / PM₁₀ and PM_{2.5}) with Grimm EDM180 technology.

In 2022, during the operation phase, it was decided to reinforce control through a hybrid air quality monitoring network consisting of 3 mobile stations Grimm EDM180 for measuring PMs and 3 Kunak AIR Pro stations for measuring polluting gases and PMs, both systems powered by solar panels to allow measurement at points of interest without the need for a connection to the electricity grid.

The objective was to measure, control, and analyze in real-time the environmental impact of mining and the activities carried out in the mine at different critical points, using the information as a tool for decision-making that continuously improves the air quality in the area.

To manage the generated information, it was necessary to have software that integrates the data from all the network's stations, control the equipment and systems' functioning, and perform advanced data analysis for decision-making. For this reason, it was decided to implement the Kunak AIR Cloud software.





Kunak AIR Cloud has made possible to integrate data from both reference instruments and the sensor network on a single platform.



Solution

Deployment of a hybrid air quality monitoring network consisting of 3 Kunak AIR Pro stations for the measurement of gases (NO_2 , SO_2 , CO and O_3) and PMs (PM_{10} , PM_4 , $\text{PM}_{2.5}$ and PM_1) and 3 Grimm EDM180 mobile solar stations for PMs (PM_{10} and $\text{PM}_{2.5}$), with total autonomy thanks to the power supply by solar panels, which allow control at critical points and mining activities, as well as the analysis of the information for the implementation of effective practices for the reduction of emissions in the activities.

Results

Thanks to the deployed network, it is possible to carry out perimeter monitoring throughout the mine and thus protect the inhabitants of nearby communities and the mine workers from potential pollution episodes.

In addition, thanks to the Kunak AIR Cloud platform, the data collected by both fixed stations and Kunak AIR Pro stations and meteorological probes are integrated, allowing advanced analysis of the behaviour of the different pollutants and, together with the system of alerts for exceeding thresholds, to make early decisions.

It also makes it possible to evaluate the efficiency of pollution control systems such as irrigation or area coverage.

Benefits

- Expansion of the environmental monitoring network.
- Integration of data from reference instruments and sensor network into Kunak AIR Cloud.
- Monitoring of emissions during mine operation.
- Real-time information on concentrations for decision-making.
- Monitoring PM concentrations in surrounding communities.
- Evaluation of the efficiency of the control systems in place.
- Monitoring and control of supplier emissions during the loading and unloading of materials.

